

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Assignment # 11

Date Given: November 05, 2025

Date Due: November 14, 2025
Total Points: 10

1. Give an example of a sequence of random variables $(X_n)_{n \geq 1}$ and a random variable X , such that, $X_n \xrightarrow{d} X$ but X_n does not converge to X in probability.
2. Show that if $c \in \mathbb{R}$ is a constant, then $X_n \xrightarrow{d} c$ implies $X_n \xrightarrow{\mathbf{P}} c$.
3. Suppose $(X_n)_{n \geq 1}$ be a sequence of continuous random variables and X is another continuous random variable, such that, $X_n \xrightarrow{d} X$. Then show that $X_n^2 \xrightarrow{d} X^2$.
4. Let $0 < p_n < 1$ be such that $np_n \rightarrow \lambda > 0$. Let $X_n \sim \text{Binomial}(n, p_n)$ and $X \sim \text{Poisson}(\lambda)$. Show that $X_n \xrightarrow{d} X$. Give all details.